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ENVELOPE 1 · TECHNICAL PROPOSAL

Supply, Installation, Commissioning and Maintenance of High-Energy Cargo X-Ray Inspection Systems

Port of Al-Noor · Tender NGPA/T/2026/014

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Submitted to	Northern Gulf Ports Authority, Tender Committee Secretariat
Bidder	**Meridian Scan Technologies Pte. Ltd.** (Singapore)
Our reference	MST-MEA-26-0914-T
Date	11 December 2026
Contact	Daniel Koh, Regional Sales Director, MEA · tenders@meridianscan.example · +65 0000 0000

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Form F-1 — Bid submission letter (technical)

MST-MEA-26-0914-T · 11 December 2026

To: The Chair, Tender Committee, Northern Gulf Ports Authority

Meridian Scan Technologies is pleased to submit its technical proposal for tender NGPA/T/2026/014. We have examined the RFP, its Annexes and **Addendum No. 1**, which we acknowledge, and we offer the Meridian HE-6 inspection systems and the Meridian ClearView image platform in accordance with these documents.

Meridian is the OEM of the systems offered. This envelope contains no prices (RFP §2.4.2). We confirm a bid validity of 120 days from the closing date of 14 December 2026 and have signed the Authority's integrity pledge.

For and on behalf of Meridian Scan Technologies Pte. Ltd.
Daniel Koh, Regional Sales Director, MEA (under power of attorney)

Date: 11 December 2026
Company seal

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1. About Meridian

Meridian Scan Technologies was founded in 2011 and designs X-ray inspection systems and the software that connects them. Meridian has installed 58 high-energy cargo systems in 19 countries and manages a fleet of more than 400 inspection systems through its ClearView platform. Production takes place in our Singapore facility; final integration and radiation testing for the Middle East are performed in our Jebel facility.

Form F-4 — References

Customer / site	Systems	Year	Climate	Reference contact
Tamarind Coast Port, Container Terminal	2 × HE-6 drive-through	2020	Mean summer max 42 °C, RH 80 %	Chief of Customs Technology
Northern Corridor rail-road border post	2 × HE-6, 1 × HE-6R	2022	Mean summer max 38 °C, RH 45 %	Border Agency Programme Director
Port of Velar, Free Zone gate	1 × HE-6	2023	Mean summer max 33 °C, RH 75 %	Free Zone Security Manager
Island Logistics Hub	3 × HE-6 with ClearView Cloud analysis	2025	Mean summer max 31 °C, RH 82 %	Head of Inspection Operations

2. Proposed solution

Our solution combines four high-energy systems with **Meridian ClearView**, a single platform for image review, workload routing, equipment monitoring and AI-assisted detection (ThreatSense).

Site	Model	Energy	View	Penetration	Throughput
T1 Gate A	HE-6 Drive	6 MeV dual-energy	Top	310 mm	190 trucks/h
T1 Gate B	HE-6 Drive	6 MeV dual-energy	Top	310 mm	190 trucks/h
T2 main gate	HE-6 Drive	6 MeV dual-energy	Top	310 mm	190 trucks/h
Relocatable	HE-6R	4.5/6 MeV interlaced dual-energy	Top	290 mm	140 trucks/h

The HE-6 family is single-view by design: our image chain uses a high-resolution top-view detector (2.5 mm pixel pitch) which, in our experience, gives analysts the information they need while reducing the controlled-area footprint and the cost of shielding.

3. Imaging and throughput

Parameter	HE-6 Drive (T1, T2)	HE-6R (relocatable)
Steel penetration (ANSI N42.46-2008)	310 mm	290 mm
Wire detection behind 100 mm steel	1.0 mm Cu	1.0 mm Cu
Contrast sensitivity	1.0 %	1.0 %
Material discrimination	4 classes	4 classes

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Scan speed	5–15 km/h	5–10 km/h
Tunnel (H × W)	4.7 m × 3.1 m	4.6 m × 3.0 m

Throughput is stated as a sustained rate with the driver in the cab and 12 m spacing between trucks. All lanes accept 45' high-cube containers.

4. Radiation safety, site fit and power

The primary beam is interlocked with a cab-detection light curtain and two inductive loops: it switches on only after the cab has fully passed the beam plane. As an option, the cab can be scanned by a separate low-dose source at **0.2 µSv per pass**. The annual dose at the controlled-area boundary is below 1 mSv at full throughput (calculation MST-RP-0914). Emergency stops, interlocks, beam-on lights, sirens and bilingual signage are standard.

Site	Plot	Controlled area	Demand incl. HVAC	Supply available
T1 Gate A / B	95 × 36 m	82 × 28 m	196 kVA	400 kVA
T2 main gate	70 × 28 m	69 × 27.5 m	196 kVA	250 kVA
Relocatable, T1 pad	80 × 32 m	70 × 30 m	120 kVA	160 kVA
Relocatable, T2 pad	60 × 30 m	58 × 29 m	120 kVA	100 kVA + 150 kVA generator (Meridian supply)

Radiation Licensing Support Plan (Addendum No. 1, A4). Meridian's radiation protection team prepares the shielding calculations, radiation protection programme and licence application for each system for submission by the Employer, coordinates the regulator's inspections before each SAT, and keeps a radiation protection officer on site until all licences are granted. Application for the T1 Gate A system is planned in month 2, the others in month 4.

5. Environmental performance and UPS

Operating range 0 °C to +50 °C, humidity up to 95 % non-condensing; outdoor cabinets IP54 with closed-loop cooling; structural steel coated to ISO 12944 C5-M (high durability); wind survival 120 km/h. Power supply 400 V ±10 %, 3-phase, 50 Hz. UPS autonomy: **60 minutes** for control systems and servers, **30 minutes** for workstations.

6. Meridian ClearView platform

Deployment. ClearView image servers, the database and the analysis applications are installed on-premise at the Customs Data Center (CIAC) with edge servers at each terminal. All scan images, analyst annotations and inspection results are stored and processed at the CDC and never leave the Employer's network.

Licensing and updates. ClearView licences are managed by the **Meridian Cloud Gateway**, a virtual appliance installed in the Employer's DMZ. The gateway synchronises licence entitlements with Meridian's licensing service

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every 90 days over an outbound HTTPS connection and downloads digitally signed software and ThreatSense model updates, which the Employer's administrators approve before installation. If the connection is unavailable, a 30-day grace period applies before analysis features are restricted.

Equipment health monitoring. Meridian Insight sends equipment health data every hour (component temperatures, error codes, dose-rate logs, uptime counters; no images or cargo data) through the gateway to Meridian's monitoring service hosted in the EU, allowing our support centre to plan preventive maintenance and pre-position spare parts.

Storage. 1,170,000 scans/year × 45 MB × 5 years = 263.25 TB; with 20 % headroom 315.9 TB. We offer **320 TB usable** (erasure-coded object storage, 3 nodes). Retention is 5 years online; export in DICOS (NEMA IIC 1).

Workload routing distributes scans across CIAC and terminal analysts; expected 90th-percentile scan-to-decision time is 2.8 minutes. **ThreatSense** AI highlights anomalies for undeclared goods, weapons, cigarettes, narcotics and currency; models run on the CDC GPU servers. The Employer's images are not used for training outside the CDC without written consent.

7. Integration

CRMS: bidirectional REST/JSON over HTTPS (scan requests, decisions, image links, annotations). TOS: REST adapter or UN/EDIFACT COPARN/CODECO via the ClearView Integration Bus; all middleware included. OCR: ISO 6346 container numbers with check digit (≥ 98.5 %) and Arabic/Latin licence plates (≥ 98 %), validated over 7 days during SAT.

8. Cybersecurity

- Active Directory integration over LDAPS, RBAC with 9 predefined roles, MFA for administrators (TOTP).
- Tamper-evident audit trail retained 5 years; syslog export to the Employer's SIEM.
- AES-256 encryption at rest; TLS 1.2+ in transit.
- SBOM (SPDX) per release. Cumulative security updates are released **twice a year**; critical vulnerabilities are fixed by emergency patch within 30 days.
- Independent penetration test before go-live; critical and high findings closed before acceptance.
- Remote support sessions through the Employer's VPN, approved per session and recorded.

9. Workstations and user interface

20 workstations (10 CIAC, 5 T1, 3 T2, 2 spare) per Addendum No. 1, each with two 27-inch 2560 × 1440 calibrated displays, 32 GB RAM and a dedicated GPU. The ClearView user interface is available in Arabic and English. Dates are displayed in the Gregorian calendar; Hijri display is planned for the ClearView 8 release.

10. Project schedule

Phase	Months
Design, surveys, licence applications	M1–M3
FAT in Singapore (4 Employer witnesses, travel at Meridian's cost)	M3
Civil works T1 / T2	M2–M6

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T1 Gate A installation, SAT, trial operation starts	M6
Other systems, CIAC, integration	M6–M8
PAC of all systems	M9

11. Training

Operators (40 × 5 days, on site), analysts (12 × 10 days plus 20 days coaching at the CIAC), maintenance engineers (6 × 15 days at our Singapore factory) and system administrators (2 × 5 days at the CDC). Courses are delivered in English with simultaneous Arabic interpretation at Meridian's cost, and include a train-the-trainer track. Manuals in English and Arabic.

12. Warranty and service

Warranty of 24 months from PAC per system, covering all parts, labour and travel including the X-ray source and detectors; 36 months for IT equipment. A local service centre will be established before PAC with four OEM-certified engineers (two transferred from our regional office, two recruited locally and certified at our factory) and an in-country stock of critical spares. Remote response within 2 hours through Meridian Insight alarms, on-site response within 24 hours, 24/7. Guaranteed availability of 98 % per scanner per month. Spare parts are available for 10 years after PAC. The five-year CMC covers preventive and corrective maintenance, spares including one source replacement per system if required, and ClearView and ThreatSense updates.

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Annex 1. Form F-2 — Technical compliance matrix

One row per Annex B requirement, as amended by Addendum No. 1. "Reference" points to the section of this proposal.

Ref	Requirement	Status	Offered	Reference
TB-01	Steel penetration (ANSI N42.46): T1 A/B ≥ 310 mm, T2 ≥ 300 mm, relocatable ≥ 280 mm	Comply	310 mm (T1, T2); 290 mm (relocatable)	§2–3
TB-02	Wire detection behind 100 mm steel ≤ 1.0 mm Cu	Comply	1.0 mm Cu	§3
TB-03	Dual-energy X-ray source	Comply	Dual-energy 6 MeV; interlaced 4.5/6 MeV (relocatable)	§2
TB-04	Contrast sensitivity ≤ 1.0 %	Comply	1.0 %	§3
TB-05	Material discrimination ≥ 4 classes	Comply	4 classes	§3
TB-06	Dual-view (top + side) imaging (optional for relocatable)	Not comply	Single view (top), high-resolution	§2
TB-07	Throughput, driver in cab: T1 Gate A ≥ 180/h; T1 Gate B, T2 ≥ 150/h; relocatable ≥ 120/h (A5)	Comply	190/h (T1, T2); 140/h (relocatable)	§2
TB-08	Vehicle speed during scan 5–15 km/h	Comply	5–15 km/h (relocatable 5–10)	§3
TB-09	Inspection tunnel ≥ 4.6 m × 3.0 m (H × W)	Comply	4.7 × 3.1 m; 4.6 × 3.0 m	§3
TB-10	High-energy beam never on occupied driver cab	Comply	Light curtain + loops interlock	§4
TB-11	Driver dose per pass ≤ 0.25 µSv (if cab scan offered)	Comply	0.2 µSv (optional cab scan)	§4
TB-12	Annual dose at controlled-area boundary ≤ 1 mSv	Comply	< 1 mSv/yr	§4
TB-13	Controlled area + exclusion zones fit the site plots	Comply	T2: 69 × 27.5 m	§4
TB-14	Emergency stops, interlocks, beam-on lights, audible alarms	Comply	Standard	§4
TB-15	Continuous operating ambient 0 … +50 °C	Comply	0 … +50 °C	§5
TB-16	Relative humidity ≤ 95 % non-condensing	Comply	≤ 95 %	§5
TB-17	Outdoor electronics enclosures ≥ IP54	Comply	IP54	§5
TB-18	Steel coating C5-M (ISO 12944)	Comply	C5-M	§5
TB-19	Wind survival 120 km/h	Comply	120 km/h	§5
TB-20	Supply 400 V ±10 %, 3-phase, 50 Hz	Comply	400 V, 3-ph, 50 Hz	§5

TB-21	Maximum demand per system ≤ site capacity	Comply	196 kVA; 120 kVA + generator at T2 pad	§4
TB-22	UPS autonomy, control systems and servers ≥ 60 min (A3)	Comply	60 min	§5
TB-23	UPS autonomy, workstations ≥ 30 min	Comply	30 min	§5
TB-24	On-premise hosting of all components incl. licensing, telemetry, updates	Comply	ClearView on-premise at CDC; licensing and updates via Meridian Cloud Gateway (no image data)	§6
TB-25	Data residency: in-country only	Comply	Images and results stay at CDC	§6
TB-26	Online retention of images and metadata: 5 years	Comply	5 years	§6
TB-27	Usable storage at CDC per RFP §4.6.3 (state usable TB)	Comply	320 TB usable	§6
TB-28	DICOS (NEMA IIC 1) export	Comply	DICOS export	§6
TB-29	CRMS bidirectional REST/JSON integration	Comply	REST/JSON	§7
TB-30	TOS integration: EDIFACT COPARN/CODECO or REST	Comply	REST or EDIFACT	§7
TB-31	OCR container numbers (ISO 6346) ≥ 98 %	Comply	≥ 98.5 %	§7
TB-32	OCR licence plates, Arabic + Latin ≥ 98 %	Comply	≥ 98 %	§7
TB-33	Scan-to-analyst-decision time, 90th percentile ≤ 3 min	Comply	2.8 min expected	§6
TB-34	AD via LDAPS, RBAC, MFA for administrators	Comply	LDAPS, RBAC, TOTP MFA	§8
TB-35	Tamper-evident audit log retention 5 years	Comply	5 years	§8
TB-36	Encryption AES-256 at rest / TLS ≥ 1.2 in transit	Comply	AES-256 / TLS 1.2+	§8
TB-37	Vendor remote access via Employer VPN, per session, recorded	Comply	Employer VPN per session, recorded	§8
TB-38	Operator/analyst UI in Arabic and English	Comply	Arabic + English	§9
TB-39	Hijri + Gregorian date display	Not comply	Gregorian only (Hijri planned)	§9
TB-40	Monitors: 2 × 27", ≥ 2560 × 1440 per workstation	Comply	2 × 27" 2560 × 1440	§9
TB-41	Workstations: 20 in total (A2)	Comply	20	§9
TB-42	Relocation T1 ↔ T2 incl. re-commissioning ≤ 72 h	Comply	60 hours	§2
TB-43	Availability per scanner per month ≥ 98 %	Comply	98 %	§12
TB-44	On-site response 24/7 ≤ 24 h	Comply	24 hours	§12

TB-45	Spare parts availability ≥ 10 years after PAC	Comply	10 years	§12

Annex 2. Form F-6 — Statement of deviations (technical)

The Bidder declares that this proposal contains **no deviations** from the tender documents, including Addendum No. 1.

For and on behalf of Meridian Scan Technologies Pte. Ltd.
Daniel Koh, Regional Sales Director, MEA (under power of attorney)

Date: 11 December 2026
Company seal

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